Opening Physics Data: Lessons from CTH

Prepared by Prof. Evdokiya Kostadinova, Auburn University

The challenge. With the rise of data-driven approaches to scientific discovery, open-sourcing large experimental datasets is highly desirable. However, making data open is not simply a matter of providing access to files. Data must be documented, discoverable, technically usable, and accompanied by clear terms that tell researchers how the data can be used and credited.

CTH example: tiered data access

The Compact Toroidal Hybrid (CTH) facility at Auburn University (AU) is a plasma confinement device that aims to study fusion energy. The facility has been used for scientific research for over a decade, which has resulted in a large database of experiments. Through a three-year collaboration with MIT, UW–Madison, William & Mary, and the HDF5 Group, the CTH team has been developing workflows for sharing fusion datasets according to FAIR principles (Findable, Accessible, Interoperable, and Reusable). To do this, several options were considered based on the following categories of data users: (i) CTH team and AU-affiliated collaborators, (ii) external collaborators, and (iii) public users.

The first group includes researchers who have Auburn IDs and can be given direct access to the database after agreement from the CTH leadership is obtained and appropriate training is conducted. For collaborators external to Auburn, unpublished CTH datasets can be accessed after a user agreement is reached and signed (see details on the user agreement in the next section). Finally, datasets already published in peer-reviewed journal articles can be fully open-sourced through Zenodo, assigned DOIs, and shared with users after signing standard licenses such as Creative Commons Attribution (CC BY). This approach guarantees that the open-sourced datasets have been vetted by the experimental team and allows the users to learn how to analyze or plot the data by comparing against figures in the associated published paper. Finally, this approach benefits the facility team as it gives visibility to their research.

Data format and user agreement

CTH data is currently managed in MDSPlus, which is an open-source software system that organizes experimental and simulation data into "trees," where each tree represents an experiment, a diagnostic system, or a logical grouping of signals. Within a tree, data are stored in a hierarchy of "nodes" that can hold raw signals, processed results, calibration details, or metadata. However, most data-driven analysis algorithms need tabular data formats, such as grids of rows and columns. To address this problem, we developed routines that retrieve data from the MDSPlus tree structure and convert it into HDF5 files – a hybrid architecture that blends tree-like hierarchy with high-performance tabular/array storage. Those routines were collected in a GitHub repository and can be shared with collaborators with access to the CTH database. CTH team members can also use those routines to pull selected datasets from the database and prepare them for open-sourcing through Zenodo.

Meanwhile, datasets not ready for public release need a separate controlled-sharing agreement for internal and external collaborators. CTH is funded by the Department of

Energy and is hosted by a land-grant public university. Therefore, open-sourcing data from the facility has to be performed with caution. To address this, we developed a CTH data user agreement based on MIT's Outbound Data Use Agreement developed for the Alcator-C Mod fusion facility. It identifies approved researchers and research purposes, prohibits onward distribution without written permission, and addresses security, incident notification, publication review process, attribution, termination, and data destruction or retention. The data license agreement is valid for a two-year period. The CTH user agreement aims to distinguish publicly licensed datasets from controlled-access data, make researcher and project scope clear, and keep publication, citation, security, and retention requirements aligned with the facility team needs, the home institution requirements, and the federal agency guidelines.

Academic data centers

Facility-by-facility solutions are valuable, but shared institutional capacity can make good practice repeatable. At Auburn, we are currently working on establishing the Center for Data Science Innovation (CDSI), which aims to promote data stewardship and reproducibility, including FAIR-aligned practices, metadata standards, provenance tracking, version control, reusable workflows, and research-data support. Common metadata, formats, persistent identifiers, transparent licenses, and documented access conditions make it easier to compare or combine data across experiments and to build reproducible analyses that can travel between facilities. Institutional data centers can coordinate these practices without requiring every facility to use identical infrastructure or surrender control over data that legitimately remain restricted.

© 2026 Evdokiya Kostadinova. This work is licensed under a Creative Commons Attribution 4.0 International License (CC BY 4.0), https://creativecommons.org/licenses/by/4.0/.